Multi- quadrant Operation

- For consideration of multi quadrant operation of drives, it is useful to establish switchable conventions about the signs of torque and speed.
- Motor speed is considered positive when rotating in the forward direction in loads involving up and down motions, the speed of the motor which causes upward motion is considered forward motion and as positive.
- For reversible drives, forward speed is chosen arbitrarily.
- Positive motor torque is defined as the torque that produces acceleration or the positive rate of change of speed in the forward direction.
- Positive load torque is opposite in direction to the positive motor torque $T=T_L+Jd\omega/dt$.
- Motor torque is considered negative if it produces deceleration.
- A motor operates in two modes- motoring and braking.
- In motoring mode, it converts electrical energy to mechanical energy, which supports its motion. In braking, it works as a generator converting mechanical energy to electrical energy, and opposes the motion.

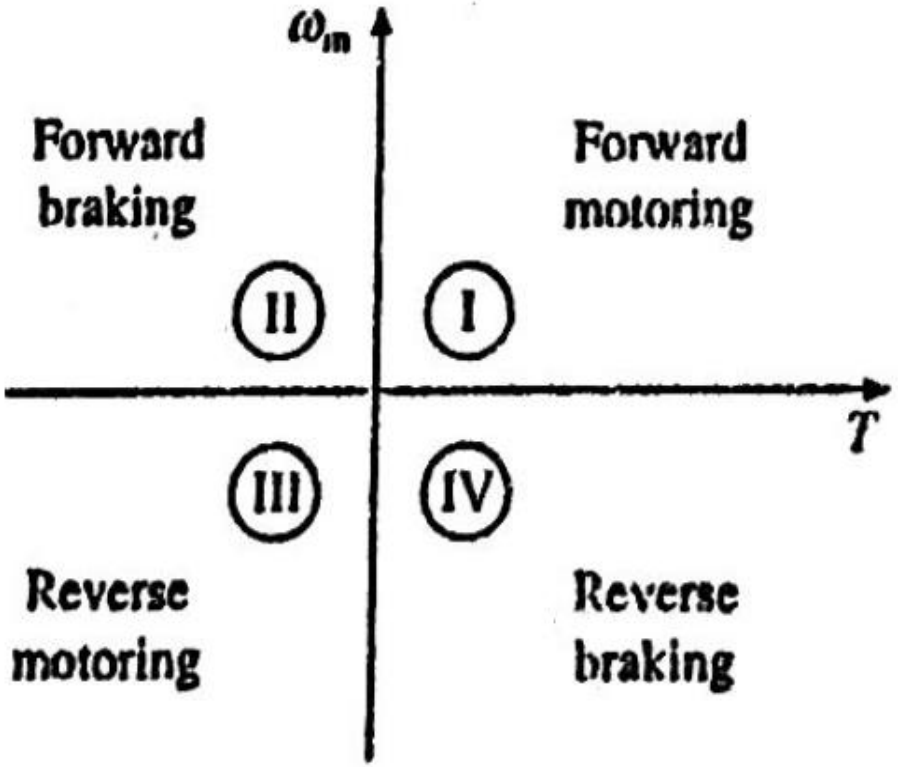


Figure 1 Multiquadrant-operation of drives

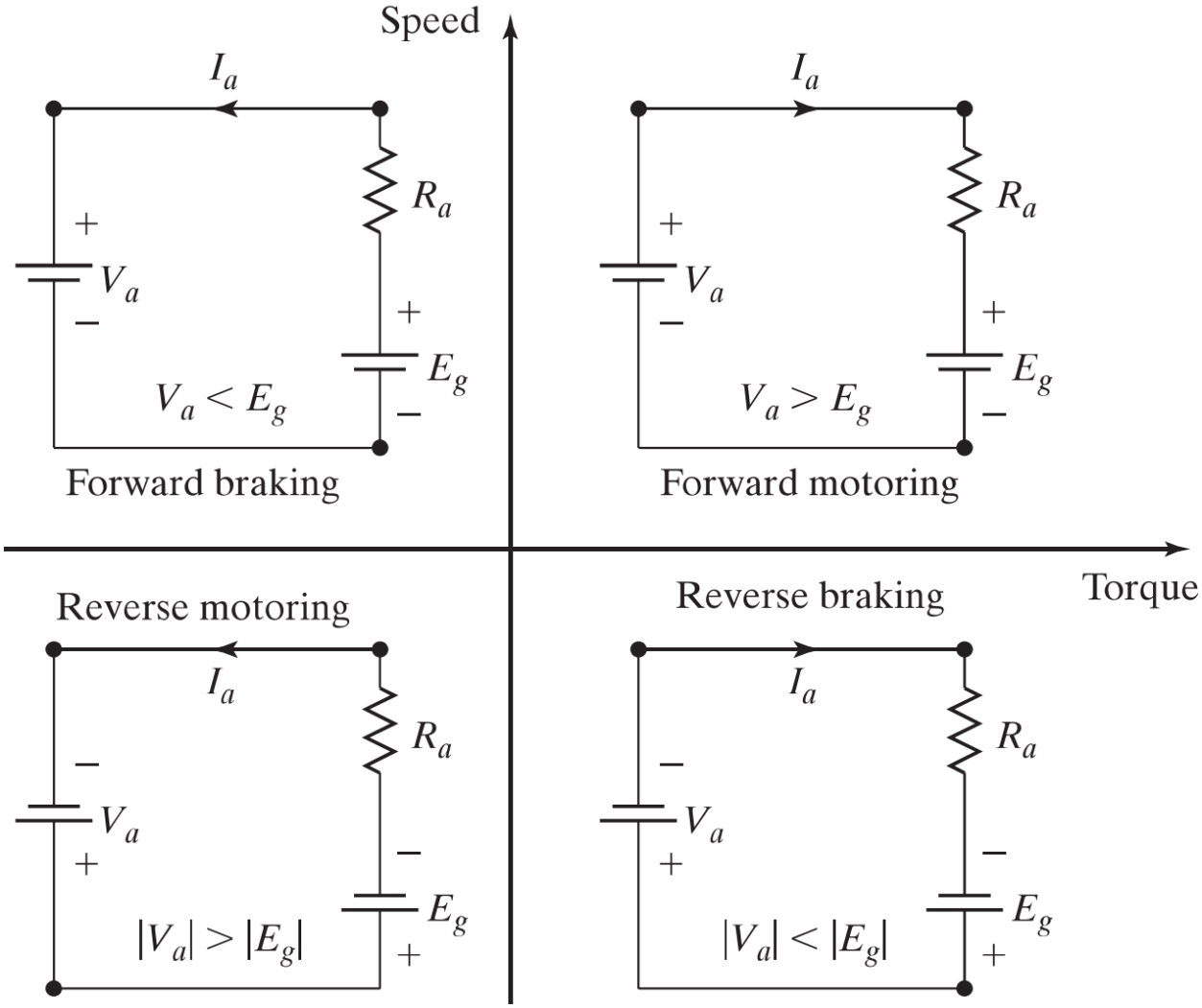
- The motor can provide motoring and braking operations for both forward and reverse directions.
- The torque and speed coordinates for both forward and reverse motions are shown in the above figure.
- Power developed by a motor is given by the product of speed and torque.

1- In quadrant I developed power is positive and called forward motoring mode.

2- In quadrant II, the developed power is negative the machine works under braking opposing the motion and this is known as forward braking mode.

3- Similarly, quadrant III known as reverses motoring. And quadrant IV, known as reverses braking.

Conditions for four quadrants



Four-quadrant operation of a hoist

- Consider the operation of a hoist in four quadrants as shown in below figure 2:
- A Hoist consists of a rope wound on a drum coupled to the motor shaft.
- One end of the rope is tied to a cage which is used to transport objects from one level to another level where the other end of the rope has a counterweight.
- The weight of the counterweight is chosen to be higher than the weight of an empty cage but lower than a fully loaded cage.
- Load torque has been shown to be constant and independent of speed, by neglecting forces due to friction and windage compared to those due to gravity.
- Load torque line T_L in quadrants I&IV represents speed-torque characteristics for the loaded hoist. This torque is the difference of torques due to the loaded hoist and counterweight.
- The load torque line T_{L2} in quadrants II &III is the speed- torque characteristic for an empty hoist. This torque is the difference of torques due to the counterweight and the empty hoist. And it is always negative.

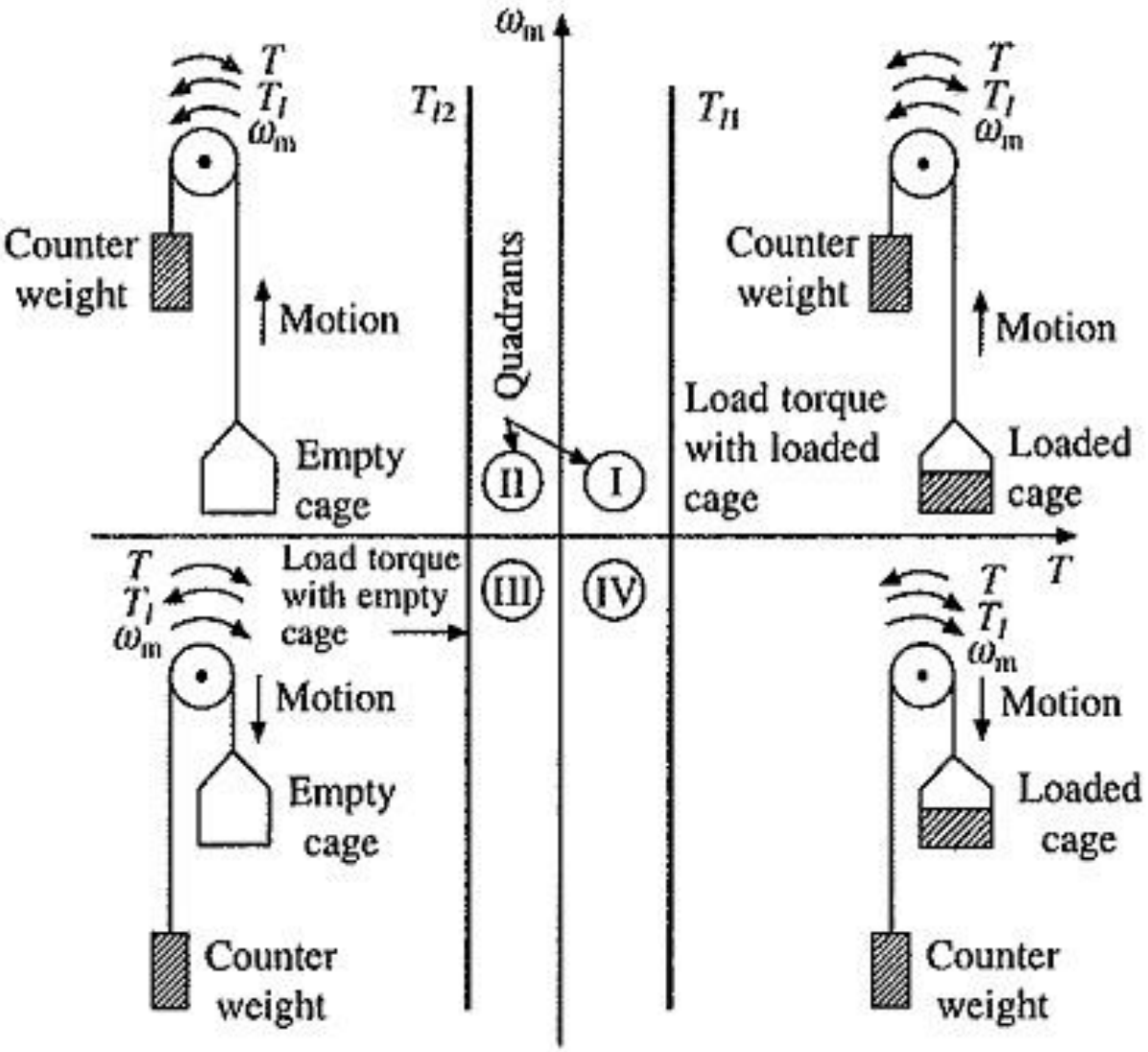


Fig. 2.3 Four quadrant operation of a motor driving a hoist load

Quadrant I operation

- The Quadrant I operation of the hoist requires

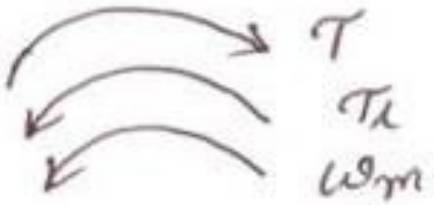
the movement of the cages upwards which corresponds to the speed (anticlockwise direction)



- This motion will obtained if the motor produces positive torque in an anticlockwise direction equal to the magnitude of the load torque (t_L).
- Since the developed motor power is positive, this is a forward motoring operation.
- motoring mode occurs at the region between no-load and short circuit points in the first quadrant. Motor parameters are:

$0<\omega<\omega_O$, $0< T <T_{sc}$, $P_2=\omega T >0$, $P_1=P_2+ \sum\Delta P$, $0< I_m <I_{sc}$, $\sum\Delta P=I_m^2 R_m$- copper losses. $0 \le \eta \le \eta_n$. At nominal load, the motor will get the best efficiency which equals the nominal value. It is not recommended to increase the motor current over the nominal current (overloading) due to the heat increasing in the armature windings.

Quadrant II operation



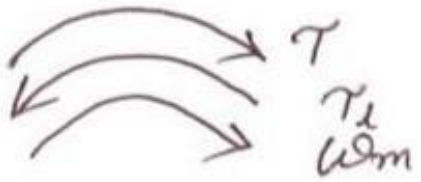
- Operation in quadrant II is obtained when

an empty cage is moved up.

Since a counterweight is heavier than an empty cage it is able to pull it up. In order to limit the speed within a safe value, the motor must produce braking torque equal to t_L in a clockwise direction (negative direction).

- Since speed is positive and developed power negative it is forward braking operation (regenerative).
- Regenerative forward braking occurs in the second quadrant when the motor speed is superior to the no-load speed. Machine works as a generator parallel to the source because the motor reverses its torque. Motor parameters are: $\omega_o<\omega$, $T < 0$, $P_2=\omega T<0$, $-P_1=-P_2+ \sum\Delta P$ which means that $P_2= P_1+ \sum\Delta P$, I_m reverses its direction, losses are electrical (heating in the copper windings) and friction.

Quadrant III operation



- Operation in quadrant III is obtained when an empty cage is lowered.
- Since an empty cage has less weight than a counterweight, the motor should produce a torque in a clockwise direction. To meet load torque T_L.

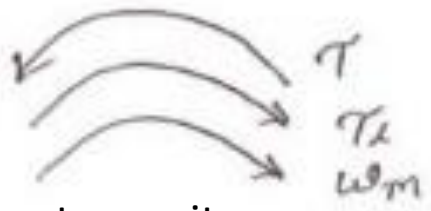
Since speed is negative and developed power is positive this is reverse motor operation.

- This motoring mode occurs at the region between no-load and short circuit points in the third quadrant. Motor parameters are:

$\omega<0$, $T < 0$, $P_2=\omega T>0$, $P_1=P_2+ \sum\Delta P$, $I_m <0$, $\sum\Delta P=I_m^2 R_m$-copper losses. $0 \le \eta \le \eta_n$. At nominal load, the motor will get the best efficienscy which equals the nominal value. It is not recommended to increase the motor current over the nominal current (overloading) due to the heat increasing in the armature windings.

Quadrant IV operation

- Quadrant IV is obtained when a loaded cage



is lowered since the weight of a loaded cage

is higher than that of a counterweight, it is able to come down due to gravity.

- In order to limit the speed of the cage within a safe value, the motor must produce a positive torque T equal to t_L in an anti-clockwise direction.
- As both power and speed are negative, the drive is operating in reverse braking mode .

Reverse braking occurs in the fourth quadrant when the motor speed is superior to the no-load speed of fourth quadrant $|\omega| > |\omega_0|$. $T > 0$, $P_2=\omega T<0$, $-P_1=-P_2+ \sum\Delta P$ which means that $P_2= P_1+ \sum\Delta P$, I_m reverses its direction relative to the second quadrant (speed is clock –wise). losses are electrical (heating in the copper windings) and friction.
Machine works as a generator parallel to the source because the motor reverses its torque relative to the second quadrant (speed is clock –wise).